



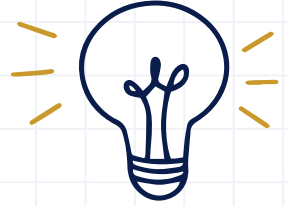
Temple Navigation System for Exploration Using Hash Table, Tree, Graph, and Dijkstra's

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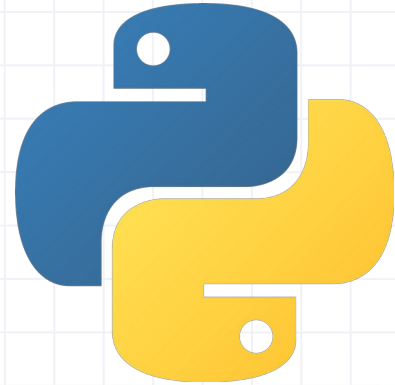
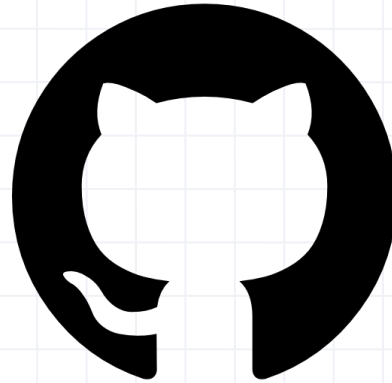
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**Conclusions &
Resources**

Tools we use:

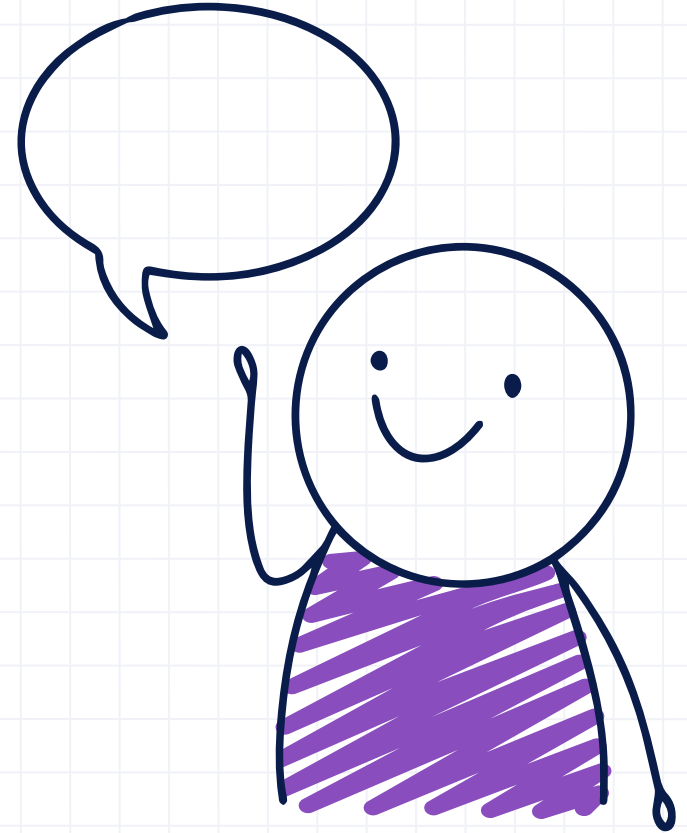


draw.io



01

Introduction PS



Problem



Visitors often have difficulty finding temples and planning the best route because the Angkor Heritage Site is very large and complex



Temple information is scattered, and there is no simple tool to show the shortest route, travel distance, travel time, or nearby temples



As a result, many tourists, especially first-time and independent travelers, may miss hidden Khmer temples and have to rely on tour guides instead of exploring on their own

Solutions



- Search for any temple quickly by entering its name and viewing detailed information in seconds.
- Find the best route between two temples with the shortest distance and estimated travel time.
- Discover nearby and related temples through recommendations, making it easier to explore more of Angkor.
- Access all temple information in one place, including temple details, routes, travel distance, and recommendations, without searching multiple sources.
- Explore Angkor independently by planning trips and finding temples on your own, reducing the need to rely on tour guides.

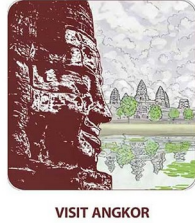
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— Literature Review



**Why Hash Table, Tree, Graph, and Dijkstra's Algorithm
are appropriate choices ?????**

supported by previous research



APSARA's Visit Angkor app provides temple information, maps, and travel guides.

However, it does not explain how route planning algorithms work

This project focuses on implementing the core data structure and algorithms for temple navigation

Hash table

Hash tables provide fast searches by name or ID
This project uses a hash table for quick temple lookup

Tree

Organize information into categories
This project groups temples by type and recommends related temples

Dijkstra's Algorithm

Finds the shortest route between two temples

Graph

Show connections between temples when calculated weight by dijkstra's algorithm

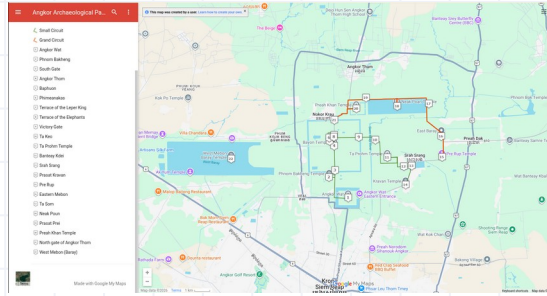
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Research Design & System Design

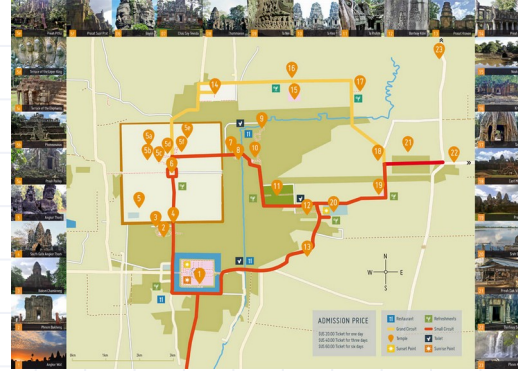


Research Design

Why 15 temples? – How were these temples selected? – Why are these distances (weights) assigned between the temples?



<https://www.google.com/maps/d/viewer?mid=1jgdy0sU3orLVU6ozXbwewd0yAojL08hY&femb=1&ll=13.44092310377663%2C103.86084845&z=13>



<https://www.visit-angkor.org/angkor-map/>



Name of temples

15



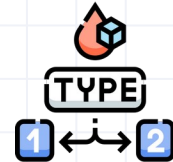
Weight between two temples

20



Information of temple

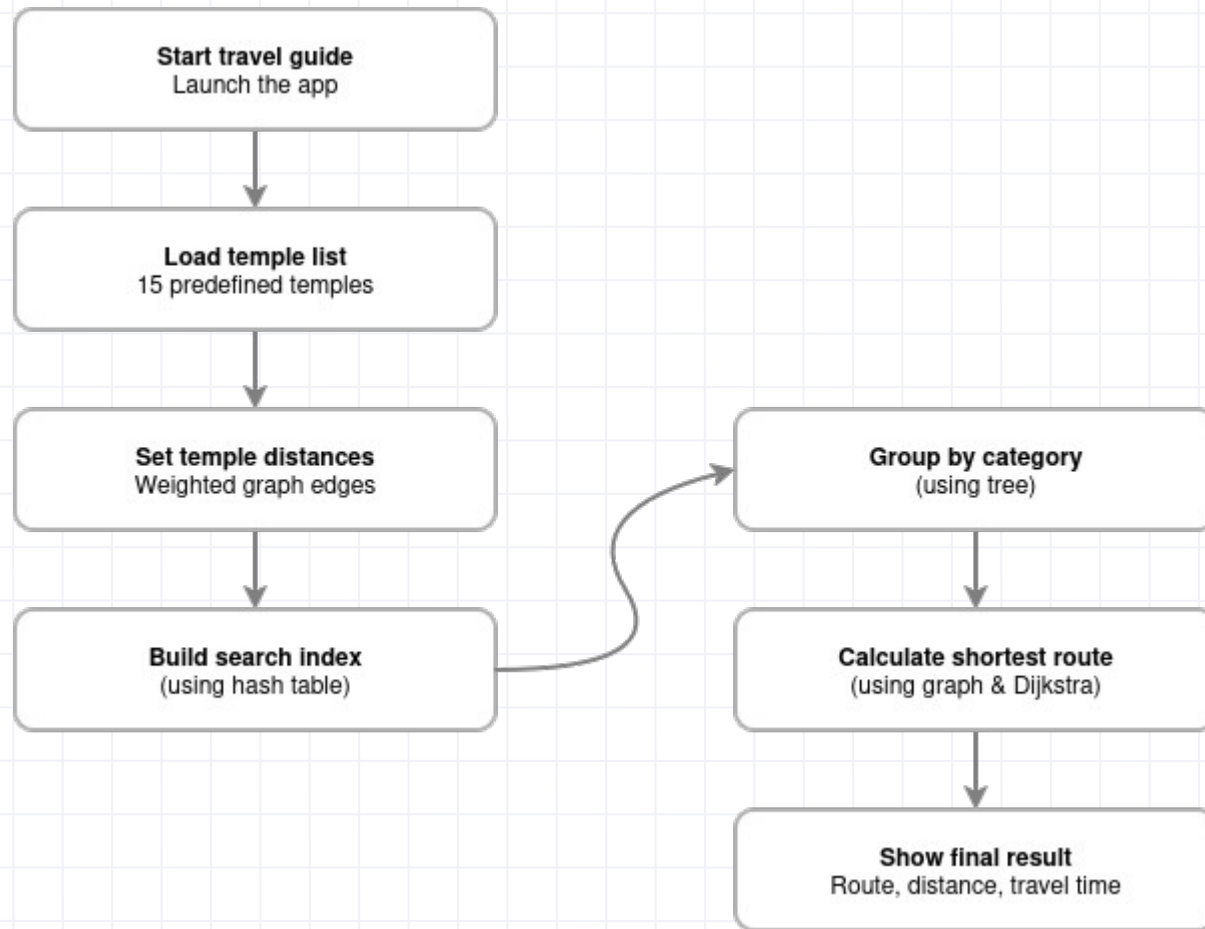
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Type of temples

4

System Design



04

Result





=====

ANGKOR TEMPLE TRAVEL GUIDE

Explore 15 temples across the park on your own.

=====

What would you like to do?

1. Find a temple
2. Plan a route between two temples
3. Explore temples by type
4. See the full temple list
5. Exit



Where are you now? (temple name or number): 1

Where do you want to go? (temple name or number): 3

Best route from Angkor Wat to Baphuon:

Angkor Wat -> Bayon -> Baphuon

Distance : 3.5 km

Estimated travel : 10 minutes (at ~20 km/h)



What type of temple do you want to explore?

1. All temples
 2. State temple
 3. Monastery
 4. Water temple
 5. Royal structure
-



Which temple are you looking for? (name or number): 1

[1] Angkor Wat

Category : State temple

Built : 12th century

Location note : South of Angkor Thom

About : The largest religious monument in the world, built as a Hindu temple and later transformed into a Buddhist site.

You might also enjoy these 6 temple(s) nearby, in the same category (State temple):

- Bayon
- Baphuon
- Phnom Bakheng
- Pre Rup
- East Mebon
- Ta Keo

DETAILS



Want to see details for the next one? (y/n):



Temple list (search by name or by its number):

1. Angkor Wat (State temple)
 2. Bayon (State temple)
 3. Baphuon (State temple)
 4. Phnom Bakheng (State temple)
 5. Pre Rup (State temple)
 6. East Mebon (State temple)
 7. Ta Keo (State temple)
 8. Ta Prohm (Monastery)
 9. Preah Khan (Monastery)
 10. Banteay Kdei (Monastery)
 11. Banteay Srei (Monastery)
 12. Neak Pean (Water temple)
 13. Srah Srang (Water temple)
 14. Terrace of the Elephants (Royal structure)
 15. Beng Mealea (Royal structure)
-



05

Conclusion & Resources

Conclusion

this project shows how data structures and algorithms can solve real-world problems. It helps visitors search for temple information, find the shortest routes, and discover more temples more easily.

Although this is an offline Python terminal application, it demonstrates a strong foundation that can be expanded into a complete temple navigation system in the future.

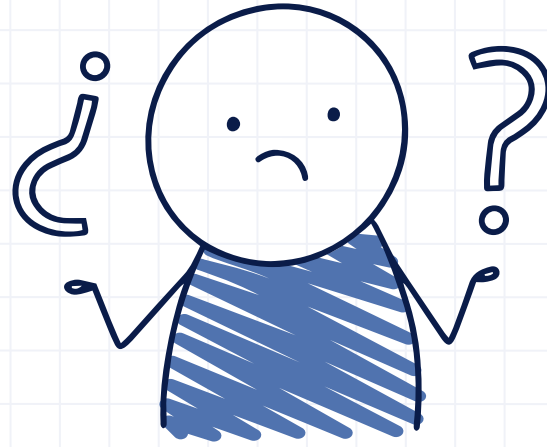
Resources

Implement python code

- <https://github.com/Hangsovoleak/angkor-heritage-site-navigation-system.git>

Data temple

- <https://www.visit-angkor.org/angkor-map/>
- <https://helloangkor.com/navigating-angkors-temples-the-small-circuit-the-grand-circuit-map-guide/>



**HAVE ANY
QUESTIONS?**

Thanks!

